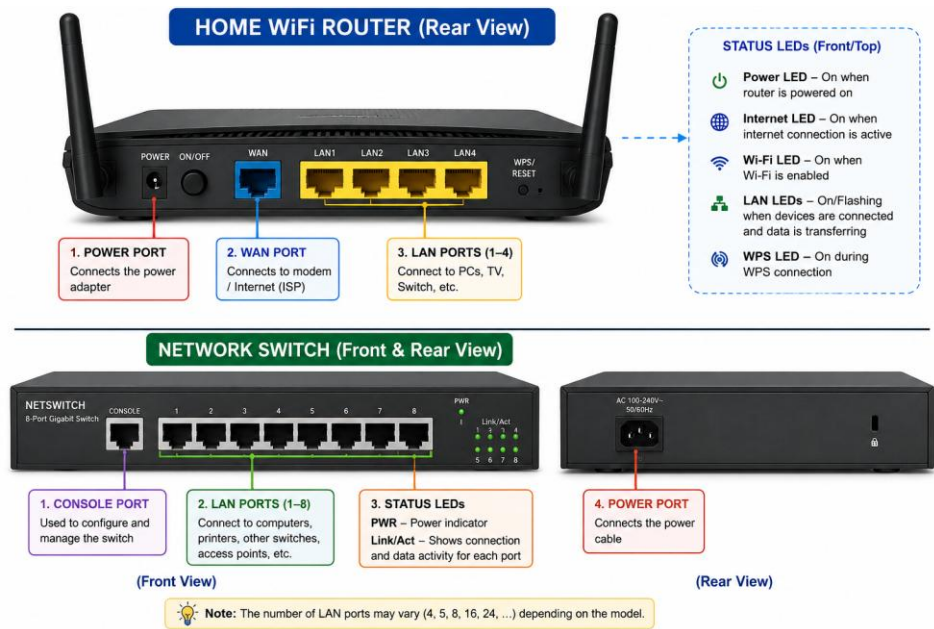


1: Diagram of a Home Wi-Fi Router and a Network Switch



2: Difference between Router & Switch

Feature	Router	Switch
Main Function	Connects different networks (LAN to Internet).	Connects devices within the same LAN.
Address Used	Uses IP addresses to route data between networks.	Uses MAC addresses to send data to the correct device.
Network Traffic	Directs traffic between different networks and the Internet.	Forwards traffic only within the local network (LAN).

3: Identify Router and Switch Components

A. Wi-Fi Router

Components

- Power Port:** Located on the back; used to connect the power adapter.
- WAN Port:** Usually blue or yellow; connects to the modem/ISP.

- **LAN Ports:** Usually 4 Ethernet ports; connect PCs, switches, smart TVs, etc.
- **LED Indicators:** Located on the front or top; show Power, Internet, Wi-Fi, and LAN activity.

Purpose

- Power Port → Supplies electricity.
- WAN Port → Receives Internet from ISP.
- LAN Ports → Connect wired devices.
- LEDs → Display device and network status.

B. Network Switch

Components

- **Power Port:** Rear side; connects the power adapter.
- **Ethernet Ports (LAN1–LAN8):** Connect computers, printers, or other switches.
- **Console Port (Managed Switch):** Used for configuration and management.
- **LED Indicators:** Show Power and Link/Activity for each Ethernet port.

Purpose

- Power Port : Powers the switch.
- Ethernet Ports : Connect wired devices.
- Console Port : Device configuration.
- LEDs : Indicate port connection and data activity.

4: Gaming Cafe

A small gaming cafe is being set up for IPL live streaming and multiplayer online games. The router is connected first to the ISP's Internet line because it receives the Internet connection, assigns IP addresses using DHCP, and provides security through NAT and a firewall.

An 8-port Gigabit switch is then connected to one of the router's LAN ports. All 8 gaming PCs are connected to the switch using Cat6 Ethernet cables. This arrangement allows every PC to communicate at high speed over the local network while sharing the same Internet connection from the router.

Router: Connects the café to the Internet, manages IP addresses, and protects the network.

Switch: Provides multiple high-speed wired connections for all gaming PCs with low latency, making it ideal for online gaming and HD IPL streaming.

